Bangladedsh University of Engineering and Technology

Project: Titumir Hall's Meal Optimization (Individual)

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Course Name: Operation Researches

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Scenario:

January 2024 months' budget is BDT 1200000. There are 480 students living in this students' dormitory and about 75% students take their meals regularly. Other 25% students do not take their meal because they have their accommodation in Dhaka or other reasons. Meals must provide at least daily essential nutrients per day. Student complain about repetitive menus with limited variety. Besides Lemon slice must be provided in every meal. Lentils is served on daily basis.

There are some dietary restrictions for some students. About 15% students are non-Muslim. Besides some food contain allergen ingredients. They must be minimized. Fresh vegetables spoil quickly due to limited storage space. Kitchen item is limited but sufficient for this number of students. Food waste have to be minimized to reduce cost and environmental impact.

Beef and mutton items are very costly. But they must be served at least 1 day per week to make variation. Due to dietary restrictions non-Muslims cannot take beef. So mutton should be provided instead of beef for them. Bagda Chingri should be introduced at least two days per months. But they contain allergen. So if any student can take Chicken instead of Bagda Chingri. According to an internal online survey of the hall, about 5% students are concern about their health. So only they will take Chicken instead of Bagda Chingri.

Ingredient and Nutritional Info [1]:

		Nutrient Per 100g ingredient			
Sl No	Ingredients	Protein	Carbs	Fat	Price/kg
1	Chicken	31	0	3.6	180
2	Egg	13	0.7	9.5	160
3	Tilapia	26	0	2.7	250
4	Salmon(Rui)	22	0	12	330
5	Climbing perch fish	22.6	14	12.5	450
6	Fresh water Catfish	18	0	7.2	610
7	Common Carp (Karfu)	24	0	8.1	310
8	Bean	8.7	23	0.5	65
9	Eggplant	0.8	8.7	0.2	25
10	White Radish	0.7	3.4	0.2	12
11	Papaya	0.5	11	0.3	80
12	Potato	2.5	21	0.1	50
13	Red amaranth	2.8	5.4	0.2	30
14	Spinach	3	3.8	0.3	35
15	Long Bean	2.8	8.3	0.4	110
16	Gourd	0.9	5.4	0	30
17	Pumpkin Leaves	3.15	2.33	0.4	50
18	Broccoli	2.4	7.2	0.4	60
19	Cabbage	1.3	5.5	0.1	30
20	Cauliflower	1.8	4.1	0.5	30
21	Soybean Oil	0	0	100	160

22	Salt	-	-	-	35
23	Turmeric Powder	-	-	-	440
24	Green Chili	-	-	-	60
25	Coriander leaves	-	-	-	35
26	Tomato	-	-	-	50
27	Onion	-	-	-	80
28	Beef	26	0	17	650
29	Mutton	25	0	21	1000
30	Bagda Chingri	23	1.5	1.7	450
31	Lentils	9	20	0.4	95
32	Lemon	1.1	9.3	0.3	120
33	Rice	2.7	28	0.3	50
34	Transportation Expenses	-	-	-	600

The green colored cells are marked as constantly daily needed.

Nutritional Requirements (Daily per student):

Nutrients	Least Daily Intake
Protein	50 g
Carbohydrate	150 g
Fat	30 g

Solution Model:

Sets:

I: Sets of ingredients type

Parameters:

a_{ij}: nutritional content of ingredient i for nutrition j

b_i: recommended daily intake of nutrient i

c_i: cost per kilogram for ingredient i

Decision Variables:

x_i = the quantity of food ingredient i (in kilogram)

Objective:

Minimizing the cost of the ingredients.

$$minimize \sum_{i=1}^{n} c_i \cdot x_i$$

Constraints:

1. Nutritional Constraints:
   $a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n \geq b_i$
2. Budget Constraints:
   $$\sum_{i=1}^{n} c_i \cdot x_i \leq Budget$$

3. Value constraints:
   They are mentioned in the excel file.

Appendix-

1. Nutrition Info are collected from Nutritionix database - https://www.nutritionix.com/